

Research Note on MathOverflow 413935

Exact relaxations, finite-temperature barriers, and a sharper proof frontier

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Status and nonclaim. This note does not prove that the limit in Question 1 exists, and it does not prove nonexistence. It contains independently checked partial theorems and exact finite computations. It must not be cited or submitted as a solution to the MathOverflow question.

Abstract

For a symmetric zero-diagonal sign matrix A , put

$$M(A) = \max_{x \in \{-1,1\}^n} \left| \frac{1}{2} x^\top A x \right|, \quad F(n) = \min_A M(A).$$

The open question is whether $F(n)/n^{3/2}$ converges. We first audit the previous bounds $1/\pi \leq \liminf \leq \limsup \leq 1/2$, the augmented cut-code identity, and one-vertex monotonicity; all survive audit. We then show that the proposed cube relaxation has the unique optimizer zero at zero temperature and at every fixed unscaled positive temperature, with a leading-order zero-temperature integrality gap. In contrast, an ellipsope relaxation has the exact thermodynamic limit $1/2$, while the defined two-covariance, edgewise-oriented Gaussian certificate has asymptotic lower-bound constant $1/\pi$. We identify the augmented cut code as a binary linear code whose dual consists of even-cardinality Eulerian subgraphs, and derive an exact signed-Eulerian MacWilliams formula. Ordinary graphons and limiting empirical spectra are proved to erase the $n^{3/2}$ functional. Exhaustive computation gives $F(2), \dots, F(10) = 1, 3, 4, 4, 5, 9, 10, 12, 13$. Exact cross-block computation shows that optimal blocks of orders 2 and 8 cannot occur together in an order-10 optimum. We also identify the one-vertex Bellman state as an exact energy-weighted projective covering radius and certify that the code of exact maximizers, and even the complete scalar partition-function curve, can be insufficient. Retaining the nonlinear arcsine term gives a trace-four stability bound and the new finite consequence $F(21) \geq 32$. We derive an exact multivertex weighted Bellman identity, a weighted entropy upper bound, and density-one control of Bellman-optimal predecessors. A complete order-nine catalogue shows that even the energy histogram plus the two-point overlap law of exact maximizers does not determine extension. The sharp remaining target is a quenched minimax free-energy limit that absorbs, rather than separately bounds, the cross-block contribution.

1 Problem, audited baseline, and scope

The question, asked by Paata Ivanisvili on MathOverflow in 2022 [1], is whether

$$\lim_{n \rightarrow \infty} \frac{F(n)}{n^{3/2}}$$

exists. Here $A_{ii} = 0$, $A_{ij} = A_{ji} = a_{ij} \in \{-1, 1\}$, and

$$Q_A(x) = \sum_{i < j} a_{ij} x_i x_j = \frac{1}{2} x^\top A x.$$

The earlier attempt was rechecked from source rather than assumed. Its central conclusions are correct:

$$\frac{1}{\pi} \leq \liminf_{n \rightarrow \infty} \frac{F(n)}{n^{3/2}} \leq \limsup_{n \rightarrow \infty} \frac{F(n)}{n^{3/2}} \leq \frac{1}{2}, \quad (1)$$

$$F(n) = \binom{n}{2} - 2\rho(D_n), \quad F(n) \leq F(n+1) \leq F(n) + n. \quad (2)$$

The lower bound uses two Gaussian sign roundings and the arcsine law. The upper bound uses symmetric Paley conference matrices and principal submatrices. The code D_n is the cut-character set augmented by its negative. A claim-by-claim audit, including two attribution errors and a factor-of-two prose correction, is in `AUDIT.md`. Nothing in (1) or (2) implies convergence.

All new statements below distinguish a proved identity from a proposed limit theorem. Exact computations guide the search but are not used as asymptotic proofs.

2 The relaxed-coupling program collapses

Let $m = \binom{n}{2}$, let $c_x = (x_i x_j)_{i < j}$, and for arbitrary real couplings define

$$G_n(a) = \max_{x \in \{-1, 1\}^n} |\langle a, c_x \rangle|.$$

Theorem 1 (Cube relaxation). *The function G_n is a norm on $\mathbb{R}^m$. Hence*

$$\min_{a \in [-1, 1]^m} G_n(a) = 0,$$

and the unique minimizer is $a = 0$.

Proof. Homogeneity and the triangle inequality follow from the maximum of absolute linear forms. If X is uniform on $\{-1, 1\}^n$, orthogonality of distinct degree-two Walsh characters gives

$$\mathbb{E} \langle a, c_X \rangle^2 = \sum_{i < j} a_{ij}^2.$$

Thus $G_n(a) = 0$ forces $a = 0$. Since the origin belongs to the cube, the last assertion follows. $\square$

In particular, any convex feasible set in the original coupling coordinates that contains every sign coupling contains their convex hull, the whole cube, and therefore contains this zero optimizer when the objective remains G_n . By the audited lower bound, its additive integrality gap is at least $n\sqrt{n-1}/\pi$, of the full target order.

The same collapse persists at positive temperature. Retain duplicated states $(x, s) \in \{-1, 1\}^n \times \{-1, 1\}$ and put

$$L_{n,t}(a) = \frac{1}{t} \log \sum_{x,s} \exp(t \langle a, s c_x \rangle), \quad t > 0.$$

Theorem 2 (Finite-temperature cube dual). *The function $L_{n,t}$ is strictly convex and has the unique cube minimizer zero, with value*

$$\min_{a \in [-1,1]^m} L_{n,t}(a) = \frac{(n+1) \log 2}{t}.$$

It has the exact dual representation

$$\max_{p \in \Delta(\{-1,1\}^n \times \{-1,1\})} \left\{ \frac{H(p)}{t} - \|\mathbb{E}_p[sc_x]\|_1 \right\}.$$

Proof. The Hessian of $L_{n,t}$ is t times the covariance matrix of sc_x under a full-support Gibbs distribution. If a linear functional of these vectors were constant, replacing s by $-s$ would make the constant zero; Walsh orthogonality would then make the functional zero. The Hessian is therefore positive definite. Symmetry makes the gradient vanish at zero, where the value is $t^{-1} \log 2^{n+1}$.

The entropy variational formula for log-sum-exp gives

$$L_{n,t}(a) = \max_p \left\{ \langle a, \mathbb{E}_p[sc_x] \rangle + \frac{H(p)}{t} \right\}.$$

Both domains are compact and convex and the displayed function is convex in a and concave in p , so Sion's theorem applies. Minimization of a linear form over the cube is minus the ℓ^1 -norm of its coefficient. $\square$

Thus the simplest probability-measure dual is exact but vacuous for the discrete problem: the uniform state measure cancels every first moment.

There is nevertheless a useful conditional rounding statement.

Proposition 3 (Variance-sensitive sign rounding). *For $a \in [-1,1]^m$, set $V(a) = \sum_e (1 - a_e^2)$. There is a sign vector $\sigma \in \{-1,1\}^m$ for which*

$$G_n(\sigma) \leq G_n(a) + \sqrt{2V(a)(n+1) \log 2} + \frac{4}{3}(n+1) \log 2.$$

Proof. Round the coordinates independently with $\mathbb{E}\sigma_e = a_e$. For fixed x , the centered error $\sum_e (\sigma_e - a_e)c_x(e)$ has variance $V(a)$ and summands bounded by two. The two-sided Bernstein bound with parameter u is at most $2e^{-u}$ beyond $\sqrt{2V(a)u} + 4u/3$. There are 2^{n-1} distinct cut characters. Taking $u = (n+1) \log 2$ makes the union failure probability at most $1/2$. For a successful outcome, add this uniform error to $G_n(a)$. $\square$

This gives $o(n^{3/2})$ loss if $V(a) = o(n^2)$, but the cube optimizer has $V(0) = m$. The obvious spherical repairs also fail. On $\|a\|_2 = \sqrt{m}$, Parseval gives $G_n(a) \geq \sqrt{m}$, attained by a one-coordinate vector. Adding $|a_e| \leq 1$ to that sphere forces every $|a_e| = 1$, recovering the original discrete problem exactly.

3 Temperature scaling and the cross-block wall

The functional proposed in the question is

$$\Phi_{n,\beta} = \min_A \frac{1}{\beta n^{3/2}} \log \sum_x e^{\beta |Q_A(x)|}.$$

For every fixed $\beta > 0$, the maximum-versus-log-sum-exp inequality gives

$$\frac{F(n)}{n^{3/2}} \leq \Phi_{n,\beta} \leq \frac{F(n)}{n^{3/2}} + \frac{\log 2}{\beta\sqrt{n}}. \quad (3)$$

Thus existence of $\lim_n \Phi_{n,\beta}$ for even one fixed positive β is equivalent to the original limit. At this normalization, fixed temperature is already asymptotically zero temperature.

A genuinely extensive mean-field functional is

$$\Psi_{n,\beta} = \min_A \frac{1}{\beta n} \log \sum_{x,s} \exp \left(\frac{\beta s Q_A(x)}{\sqrt{n}} \right).$$

It satisfies

$$\frac{F(n)}{n^{3/2}} \leq \Psi_{n,\beta} \leq \frac{F(n)}{n^{3/2}} + \frac{(n+1) \log 2}{\beta n}. \quad (4)$$

Consequently, existence of $\lim_n \Psi_{n,\beta}$ for arbitrarily large fixed β would settle the problem: the limsup-minus-liminf gap would be at most $\log 2/\beta$ for every such β .

The direct interpolation does not have the needed accuracy. Define

$$h_n(t) = \min_A \log \sum_{x,s} e^{ts Q_A(x)}.$$

Proposition 4 (Exact annealed block estimate). *For all n, k and $t > 0$,*

$$h_{n+k}(t) \leq h_n(t) + h_k(t) - \log 2 + nk \log \cosh t.$$

Proof. For a within-block matrix A , write $Z_A^s = \sum_x e^{ts Q_A(x)}$. Replacing all coefficients of the second block by their negatives swaps its two one-sided partition sums. Of the two possible alignments, one satisfies

$$Z_A^+ Z_B^+ + Z_A^- Z_B^- \leq \frac{1}{2} (Z_A^+ + Z_A^-) (Z_B^+ + Z_B^-).$$

Choose every cross coefficient independently and uniformly. For fixed x, y, s , averaging the cross exponential gives $(\cosh t)^{nk}$. Therefore the expected augmented partition sum is at most the right side above times this factor, and some deterministic cross signing is no larger. Take logarithms and infima. $\square$

At $t = \beta/\sqrt{n+k}$ and $n/(n+k) \rightarrow \alpha$, the normalized cross cost tends to

$$\frac{\beta}{2} \alpha (1 - \alpha),$$

which is nonzero for fixed β and grows in the zero-temperature limit. The direct annealed cross block is therefore a leading term, not an error.

Finally, the absolute value cannot be silently removed. If every edge from vertex 1 is positive and every other edge negative, fix $x_1 = 1$ and put $T = \sum_{i>1} x_i$. Then

$$Q_A(x) = \frac{n}{2} - \frac{(T-1)^2}{2}.$$

Hence $\max_x Q_A(x) = \lfloor n/2 \rfloor$, whereas $\max_x |Q_A(x)| = \binom{n}{2}$.

4 Covariance relaxations: two exact asymptotic limits

Let

$$\mathcal{E}_n = \{R \succeq 0 : \text{diag } R = \mathbf{1}\}$$

be the elliptope. The full arcsine expression is tautologically exact:

$$M(A) = \max_{R \in \mathcal{E}_n} \left| \frac{2}{\pi} \sum_{i < j} a_{ij} \arcsin R_{ij} \right|. \quad (5)$$

Indeed, Gaussian sign rounding makes every value on the right at most $M(A)$, while $R = xx^\top$ attains $Q_A(x)$. The nonlinearity therefore preserves the original problem rather than relaxing it.

Linearizing gives a model with a complete thermodynamic limit. Put

$$S(A) = \max_{R \in \mathcal{E}_n} \left| \frac{1}{2} \text{tr}(AR) \right|, \quad S_n = \min_A S(A).$$

Theorem 5 (Adversarial elliptope limit). *For every $n \geq 2$,*

$$S_n \geq \frac{1}{2} n \sqrt{n-1},$$

and

$$\boxed{\lim_{n \rightarrow \infty} \frac{S_n}{n^{3/2}} = \frac{1}{2}}.$$

At each symmetric conference order, equality holds in the finite bound.

Proof. Fix any sign matrix A , set $q = n - 1$, and define

$$R^\pm = \frac{1}{2} \left(I \pm \frac{A}{\sqrt{q}} \right)^2.$$

They are positive semidefinite. Since every row of A has squared norm q , their diagonals equal one. Moreover

$$R^+ - R^- = \frac{2A}{\sqrt{q}}.$$

The corresponding two linear SDP values differ by

$$\frac{1}{2} \text{tr}(A(R^+ - R^-)) = n\sqrt{q}.$$

At least one has absolute value at least half this amount.

For the reverse asymptotic inequality, if $R \in \mathcal{E}_n$, then

$$|\text{tr}(AR)| \leq \|A\|_{\text{op}} \text{tr } R = n\|A\|_{\text{op}}.$$

A symmetric conference matrix has operator norm $\sqrt{n-1}$, so it attains equality. For general n , the prime number theorem in arithmetic progressions supplies a prime $q \equiv 1 \pmod{4}$ with $q+1 \geq n$ and $q+1 = n(1+o(1))$ [9]. Take a principal n -by- n submatrix of the symmetric Paley conference matrix of order $N = q+1$. Compression does not increase its operator norm $\sqrt{N-1}$, giving $S_n \leq n\sqrt{N-1}/2 = (1/2 + o(1))n^{3/2}$. $\square$

Rank-one $R = xx^\top$ show $S(A) \geq M(A)$. The missing transfer is an asymptotically vanishing integrality gap, which is not supplied by standard Grothendieck-type constant-factor rounding [5, 6]. The gap is already strict at order six: the exact computation below gives $F(6) = 5$, whereas Theorem 5 and a conference matrix give $S_6 = 3\sqrt{5}$.

There is also a sharp barrier for the linear Gaussian-sign method. Let $\Lambda(A)$ be the maximum of

$$\sum_{i < j} a_{ij}(R_{ij}^+ - R_{ij}^-)$$

over $R^\pm \in \mathcal{E}_n$, subject to the edgewise orientations $a_{ij}(R_{ij}^+ - R_{ij}^-) \geq 0$.

Theorem 6 (Oriented covariance barrier). *For every sign matrix A ,*

$$M(A) \geq \frac{\Lambda(A)}{\pi}, \quad \Lambda(A) \geq n\sqrt{n-1}.$$

Moreover

$$\boxed{\lim_{n \rightarrow \infty} \frac{\min_A \Lambda(A)}{n^{3/2}} = 1.}$$

Proof. Gaussian sign rounding of R^+ and R^- , followed by the arcsine law, gives two expected energies. Since $\arcsin'(r) \geq 1$ and each edge difference has the prescribed orientation, their difference is at least $2\Lambda(A)/\pi$. Both expectations lie in $[-M(A), M(A)]$, proving the first inequality.

The square covariances used in Theorem 5 satisfy the orientation constraints and have objective $n\sqrt{n-1}$. Conversely,

$$\Lambda(A) = \frac{1}{2} \operatorname{tr}(A(R^+ - R^-)) \leq n\|A\|_{\text{op}}.$$

Conference matrices attain equality. The same prime-number-theorem and Paley principal-submatrix construction used in Theorem 5 gives the matching asymptotic upper bound at every order. $\square$

Thus the optimized two-covariance, edgewise-oriented linear certificate Λ defined above has adversarial asymptotic lower-bound constant exactly $1/\pi$. Improving it requires information beyond this specific linearization; the theorem does not exclude more general covariance certificates.

5 The augmented cut code and its exact dual

Encode signs as bits via $(-1)^b$. Let $j \in \mathbb{F}_2^m$ be the all-one edge word and

$$(\delta u)_{ij} = u_i + u_j.$$

Then the augmented code in (2) is

$$D_n = \{\delta u + sj : u \in \mathbb{F}_2^n, s \in \mathbb{F}_2\}.$$

Theorem 7 (Linear code and dual). *For $n \geq 3$, D_n is a binary $[m, n]$ linear code and*

$$D_n^\perp = \{H \subseteq E(K_n) : \deg_H(v) \equiv 0 \pmod{2} \text{ for all } v, |H| \equiv 0 \pmod{2}\}.$$

For $n \geq 4$, the dual is spanned by 4-cycles, has dimension $m - n$, and minimum distance four.

Proof. The cut space has dimension $n - 1$. Its words have even parity on every triangle, whereas j has odd triangle parity, so adjoining j raises the dimension by one. A graph is orthogonal to every cut exactly when every vertex has even degree, and it is orthogonal to j exactly when its number of edges is even.

The cycle space is generated by triangles through one fixed vertex. An even-weight cycle-space word is a sum of an even number of these basis triangles. The sum of two sharing a nonfixed vertex is a 4-cycle. Two that share only the fixed vertex can be paired through an intermediate basis triangle and expressed as two 4-cycles. This proves generation. A nonzero even-cardinality Eulerian graph has at least four edges, and a 4-cycle exists. $\square$

Counting cuts by the size of one shore gives

$$W_{D_n}(z) = \frac{1}{2} \sum_{k=0}^n \binom{n}{k} \left(z^{k(n-k)} + z^{m-k(n-k)} \right). \quad (6)$$

Let $a_e = (-1)^{y_e}$ and define the exact augmented partition function

$$Z_y(\beta) = \sum_{d \in D_n} \exp(\beta[m - 2 \text{wt}(y + d)]).$$

Its ground state is $M(A)$.

Theorem 8 (Signed-Eulerian MacWilliams formula). *For $n \geq 3$,*

$$Z_y(\beta) = 2^n (\cosh \beta)^m P_y(\tanh \beta),$$

where

$$P_y(t) = \sum_{H \in D_n^\perp} (-1)^{|H \cap y|} t^{|H|}.$$

For $0 \leq t < 1$, $P_y(t) > 0$, and uniformly in y ,

$$M(A) \leq \frac{1}{\beta} \log Z_y(\beta) \leq M(A) + \frac{n \log 2}{\beta}.$$

Proof. The coset MacWilliams identity is

$$W_{y+D_n}(z) = 2^{n-m} (1+z)^m \sum_{H \in D_n^\perp} (-1)^{|H \cap y|} \left(\frac{1-z}{1+z} \right)^{|H|}.$$

Multiply by $e^{\beta m}$ and substitute $z = e^{-2\beta}$. This gives the formula because $e^\beta(1 + e^{-2\beta}) = 2 \cosh \beta$ and the final ratio is $\tanh \beta$. Positivity follows because the left side is a positive partition sum. The last display compares its largest summand with a sum of $|D_n| = 2^n$ summands. $\square$

For fixed $\beta > 0$, the minimized free energy in Theorem 8, divided by $n^{3/2}$, differs from $F(n)/n^{3/2}$ by $O(n^{-1/2})$. Thus the exact unresolved term is

$$\min_y \log P_y(\tanh \beta).$$

It must cancel the explicit $m \log \cosh \beta$ term to $n^{3/2}$ accuracy. Averaging $P_y(t)$ over all y gives one and yields only

$$F(n) \leq n \sqrt{(n-1) \log 2},$$

the usual sphere/union-bound constant, weaker than $1/2$.

Puncturing also stops at the old scale. For $n \geq 4$, deleting the final star maps D_n onto D_{n-1} , and the two lifts of each word have complementary star coordinates. Therefore

$$\rho(D_{n-1}) \leq \rho(D_n) \leq \rho(D_{n-1}) + \left\lfloor \frac{n-1}{2} \right\rfloor,$$

equivalently

$$F(n-1) + ((n-1) \bmod 2) \leq F(n) \leq F(n-1) + n-1.$$

The $n = 3$ instance of these inequalities is checked directly; the fiber there has four rather than two words. Across a two-block partition, the cross lifts are complementary rank-one sign matrices. On the dual side, the $\binom{n}{2} \binom{k}{2}$ alternating cross-block 4-cycles already contribute macroscopically at mean-field scaling. Product bounds that take absolute values discard the cancellation needed here.

6 High-temperature data do not truncate uniformly

Let X be uniform on $\{-1, 1\}^n$, let $Y = Q_A(X)$, and use the symmetrized partition function $\mathbb{E} \cosh(tY)$, equivalently the augmented state sY . Direct Walsh counting gives

$$\mathbb{E} Y^2 = m, \quad \mathbb{E} Y^4 = 3m^2 - 2m + 24 \sum_{C_4} \prod_{e \in C_4} a_e, \quad (7)$$

where the sum is over undirected 4-cycles. Indeed, a monomial in the fourth power survives exactly when every vertex has even incidence. Besides one edge four times and two edges twice, the only case is a simple 4-cycle, with multinomial factor 24.

Equivalently,

$$\kappa_4(Y) = 3 \operatorname{tr}(A^4) - 2n(n-1)(3n-4), \quad \operatorname{tr}(A^4) = n(n-1)^2 + \sum_{i \neq j} (A_{ij}^2)^2. \quad (8)$$

The second identity is $\operatorname{tr}(A^4) = \sum_{ij} (A_{ij}^2)^2$; the first follows by separating the closed walks with four distinct vertices in (7).

The expansion

$$\log \mathbb{E} \cosh(tY) = \frac{mt^2}{2} + \frac{\kappa_4(Y)t^4}{24} + O(t^6)$$

shows that, for each fixed n , all sufficiently high-temperature minimizers of this symmetrized partition function minimize $\operatorname{tr}(A^4)$. The symmetrization matters: the one-sided partition can have a signing-dependent cubic triangle term. Spectral Cauchy–Schwarz gives

$$\operatorname{tr}(A^4) \geq \frac{\operatorname{tr}(A^2)^2}{n} = n(n-1)^2.$$

For a Paley principal submatrix of order n inside a conference matrix of order $N = n(1 + o(1))$,

$$\operatorname{tr}(A^4) \leq \|A\|_{\text{op}}^2 \operatorname{tr}(A^2) \leq (N-1)n(n-1).$$

This gives the reverse asymptotic inequality, so

$$\lim_n \frac{\min_A \operatorname{tr}(A^4)}{n^3} = 1, \quad \lim_n \frac{\min_A \kappa_4(Y)}{n^3} = -3.$$

This fourth-order limit does not determine the next term. Exhausting all root-normalized order-seven signings gives

$$\min_A \text{tr}(A^4) = 342,$$

while the trace-minimizing signings have three distinct exact sixth moments,

$$50,781, \quad 53,661, \quad 59,421.$$

Thus fourth-order data alone are insufficient. No uniform remainder theorem connecting this high-temperature expansion to the required ground-state regime was obtained.

7 Ordinary graphons and empirical spectra lose the functional

For a signed matrix A , let W_A be its step graphon and normalize

$$\|W_A\|_{\square} = \frac{1}{n^2} \max_{S, T \subseteq [n]} \left| \sum_{i \in S, j \in T} a_{ij} \right|.$$

Theorem 9 (Cut-limit obstruction). *If $M(A_n) = O(n^{3/2})$, then $W_{A_n} \rightarrow 0$ in cut norm. There are two deterministic sequences satisfying this hypothesis and having the same zero graphon limit, while their normalized Boolean maxima are separated by at least $3/2$.*

Proof. For $x, y \in \{-1, 1\}^n$, split the coordinates according to whether $x_i = y_i$. Writing $z^+, z^- \in \{-1, 0, 1\}^n$ for the corresponding restrictions of y , symmetry of A cancels cross terms and gives

$$x^T A y = 2Q_A(z^+) - 2Q_A(z^-).$$

A multilinear quadratic form attains its maximum absolute value on $[-1, 1]^n$ at a Boolean vertex, so every sign bilinear form is bounded by $4M(A)$. Expressing two indicator vectors as affine combinations of sign vectors then gives

$$\|W_A\|_{\square} \leq \frac{4M(A)}{n^2}.$$

Take an infinite sequence of Paley conference matrices C_N . Their normalized Boolean maxima have limsup at most $1/2$, and their graphons tend to zero. Choose $k = \lfloor 2N^{3/4} \rfloor$ vertices and change every coefficient inside that set to $+1$, obtaining B_N . At most $\binom{k}{2} = O(N^{3/2})$ edges change, so $M(B_N) = O(N^{3/2})$ and the graphon perturbation tends to zero in L^1 . Fix the planted spins to $+1$ and average all outside spins independently. Every cross and outside term has mean zero, while the planted block contributes $\binom{k}{2}$. Some outside assignment therefore has at least that energy, and

$$\liminf_N \frac{M(B_N)}{N^{3/2}} \geq 2.$$

The separation from the conference limsup is at least $3/2$. □

The usual dense graph topology therefore collapses every target-scale near-optimizer to the same point. This is not repaired by keeping only a limiting empirical spectrum. At dense scaling, every empirical law of A/n tends to δ_0 , since its average squared eigenvalue is $(n-1)/n^2$. At fluctuation scaling there is a stronger obstruction.

Proposition 10 (Empirical-spectral obstruction). *For every fixed $d > 1/2$, there are conference matrices C_N and sign matrices B_N such that*

$$W_2(\mu_{B_N/\sqrt{N}}, \mu_{C_N/\sqrt{N}}) \longrightarrow 0,$$

but

$$\liminf_N \frac{M(B_N)}{N^{3/2}} \geq d, \quad \limsup_N \frac{M(C_N)}{N^{3/2}} \leq \frac{1}{2}.$$

Proof. Let $q_N = Q_{C_N}(\mathbf{1})$. The spectral bound gives $|q_N| \leq N\sqrt{N-1}/2$. Flip negative coefficients to positive until the all-one energy is at least $dN^{3/2}$. This needs $O(N^{3/2})$ edges, and there are $\Theta(N^2)$ available. Therefore $\|B_N - C_N\|_F^2 = O(N^{3/2})$. Hoffman–Wielandt implies

$$W_2^2(\mu_{B_N/\sqrt{N}}, \mu_{C_N/\sqrt{N}}) \leq \frac{\|B_N - C_N\|_F^2}{N^2} = O(N^{-1/2}).$$

The energy inequalities follow from the construction and the conference spectral bound. $\square$

This excludes dense graphons and limiting empirical spectral distributions, not spectral edges, eigenvectors, or a richer Gaussian-process limit.

For comparison, iid signings are rigorously worse than the conference upper constant.

Proposition 11 (Greedy energy of an iid signing). *On one infinite iid Rademacher edge array A_n , almost surely*

$$\liminf_n \frac{M(A_n)}{n^{3/2}} \geq \frac{2}{3} \sqrt{\frac{2}{\pi}} = 0.531923 \dots > \frac{1}{2}.$$

Proof. Choose spins sequentially by

$$x_1 = 1, \quad x_j = \operatorname{sgn} \sum_{i < j} a_{ij} x_i.$$

The resulting energy is

$$Q_{A_n}(x) = \sum_{j=2}^n \left| \sum_{i < j} a_{ij} x_i \right| = \sum_{k=1}^{n-1} Y_k.$$

Before column j is exposed, the earlier spins are measurable with respect to earlier columns. Multiplying the fresh iid column signs by those predictable spins preserves their iid Rademacher law. Hence the Y_k are independent and Y_k has the law of $|\varepsilon_1 + \dots + \varepsilon_k|$.

The central limit theorem and uniform integrability give $\mathbb{E}Y_k/\sqrt{k} \rightarrow \sqrt{2/\pi}$. A Riemann-sum argument therefore yields

$$\frac{1}{n^{3/2}} \sum_{k < n} \mathbb{E}Y_k \longrightarrow \sqrt{\frac{2}{\pi}} \int_0^1 \sqrt{t} dt = \frac{2}{3} \sqrt{\frac{2}{\pi}}.$$

Changing one input sign changes Y_k by at most two, so its centered moment generating function is bounded by $\exp(\lambda^2 k/2)$. Independence gives an exponential tail $2e^{-c_\varepsilon n}$ for deviations of $\sum_{k < n} Y_k$ by $\varepsilon n^{3/2}$. This is summable; Borel–Cantelli completes the proof. $\square$

This is not a lower bound on $F(n)$; it shows that typical iid signings are not asymptotically optimal.

8 Exact computation through order ten

Switching makes every edge incident with vertex zero positive. The remaining negative coefficients form a graph on $n - 1$ vertices, and vertex permutations reduce the search to one unlabeled graph in each isomorphism class. For a residual negative-edge graph G and a negative-spin set S not containing the root, the exact energy is

$$\binom{n}{2} - 2|E(G)| - 2|S|(n - |S|) + 4|\delta_G(S)|.$$

Exhausting nauty’s **geng** streams gives Table 1.

Table 1: Exact computational values.

n	2	3	4	5	6	7	8	9	10
$F(n)$	1	3	4	4	5	9	10	12	13
$F(n)/n^{3/2}$	.3536	.5774	.5000	.3578	.3402	.4860	.4419	.4444	.4111

Completeness is checked against the unlabeled-graph counts and hashes of the full graph streams. The reduced formula is checked by direct evaluation; all labeled signings are independently enumerated through $n = 6$; optional NetworkX and Z3 routes use different decoders and optimization mechanisms. Z3 independently gives UNSAT/SAT pairs certifying the displayed values for $n = 7, 8, 9$. At $n = 10$, the lower certificate is the exhaustive nauty search; the attempted Z3 lower query timed out.

The Paley conference matrix of order ten over $\mathbb{F}_9$ has exact Boolean maximum 15, strictly worse than $F(10) = 13$. The two optimal switching classes at order ten have different characteristic polynomials. Thus neither conference structure nor a unique spectrum describes all finite optimizers. These finite facts do not imply a limiting constant.

9 Finite non-heredity and an exact weighted covering state

For fixed signings A and B , define

$$J(A, B) = \min_{C \in \{-1, 1\}^{s \times k}} M \begin{pmatrix} A & C \\ C^\top & B \end{pmatrix}, \quad E(B) = J([0], B).$$

Complete cross-block enumeration over the optimal switching-permutation classes through total order ten gives, in particular,

$$\min_{M(A)=F(2), M(B)=F(8)} J(A, B) = 15 > F(10) = 13. \quad (9)$$

Hence no optimal order-10 signing contains an optimal order-8 principal submatrix. The smallest possible internal sacrifice repairs the obstruction: among the 1,044 root-normalized order-8 representatives, internal maximum 12 is necessary and sufficient for a $2 + 8$ completion of maximum 13. The same enumeration gives one-vertex extension distributions

n	distribution of $E(B)$ over optimal classes
7	$10^{\times 4}, 12^{\times 2}$
8	$12^{\times 2}$
9	$13^{\times 4}, 15^{\times 11}$
10	$17^{\times 1}, 19^{\times 1}.$

Thus the scalar $M(B)$ does not determine future composition cost.

There is, however, an exact coding-theoretic state for one-vertex extension. Let

$$\mathcal{P}_n = \{-1, 1\}^n / \{x \sim -x\}, \quad d_{\pm}([u], [v]) = \min\{d_H(u, v), d_H(u, -v)\}.$$

For an order- n signing B , put $M = M(B)$ and

$$w_B([x]) = \frac{M - |Q_B(x)|}{2}, \quad \rho_w(B) = \max_{[b] \in \mathcal{P}_n} \min_{[x] \in \mathcal{P}_n} (d_{\pm}([b], [x]) + w_B([x])).$$

The weights are nonnegative integers because every quadratic energy has the parity of $\binom{n}{2}$.

Theorem 12 (Weighted covering-radius Bellman identity). *For every order- n signing B ,*

$$E(B) = M(B) + n - 2\rho_w(B).$$

Equivalently, if

$$\delta_w(B) = \left\lfloor \frac{n}{2} \right\rfloor - \rho_w(B),$$

then

$$F(n+1) = \min_B (M(B) + (n \bmod 2) + 2\delta_w(B)),$$

where the minimum is over all order- n signings.

Proof. Let $b \in \{-1, 1\}^n$ be the incident signs and let t be the new spin. Pairing $t = 1$ and $t = -1$ gives, for every old spin vector x ,

$$\max_{t \in \{-1, 1\}} |Q_B(x) + t b \cdot x| = |Q_B(x)| + |b \cdot x|.$$

Since $|b \cdot x| = n - 2d_{\pm}([b], [x])$, the maximum over x for fixed b is

$$M + n - 2 \min_{[x] \in \mathcal{P}_n} (w_B([x]) + d_{\pm}([b], [x])).$$

Minimizing over $[b]$ proves the first identity. The equality $n - 2\lfloor n/2 \rfloor = n \bmod 2$ gives the deficit form. Finally, minimizing over all B and b enumerates every order- $(n+1)$ signing. $\square$

For every center $[b]$, some exact maximizer has projective distance at most $\lfloor n/2 \rfloor$. A configuration of weight greater than this can never attain the inner minimum. Consequently $E(B)$ depends only on the exact near-ground window

$$|Q_B(x)| \geq M(B) - 2 \left\lfloor \frac{n}{2} \right\rfloor.$$

The Bellman state has energy width at most n ; deeper configurations are rigorously irrelevant to one-vertex extension.

Let $X_B = \{[x] : |Q_B(x)| = M(B)\}$ and let $\rho_{\text{ext}}(B)$ be its ordinary covering radius in $\mathcal{P}_n$. The same proof, restricted to the maximizing configurations, gives the exact lower obstruction

$$E(B) \geq M(B) + n - 2\rho_{\text{ext}}(B), \quad \rho_w(B) \leq \rho_{\text{ext}}(B).$$

The exhaustive verifier independently recomputes both radii and the full extension values. At order nine, the four extendible classes have $(\rho_{\text{ext}}, \rho_w) = (4, 4)$, while the eleven failing classes have $(3, 3)$. The order-10 classes have radii $(3, 3)$ and $(2, 2)$, exactly explaining extension values 17 and 19. Exact

maximizers are not a complete state: all six order-7 classes have $\rho_{\text{ext}} = 3$, but the $|Q_B| = 7$ layer lowers ρ_w to 2 in precisely the two classes with $E(B) = 12$. This is a finite counterexample to any cavity rule retaining only the ground-state code. The full energy-weighted projective landscape is exact for one step.

The scalar partition function is also insufficient. The two optimal order-nine graph6 records G?qmaw and GCpbaw have the same complete projective absolute-energy histogram

$$\#\{|Q_B| = 0, 4, 8, 12\} = (60, 111, 60, 25),$$

and therefore have identical values of $\sum_x e^{t|Q_B(x)|}$ and $\sum_{x,s} e^{tsQ_B(x)}$ for every real t . Nevertheless their extremizer covering radii are 4 and 3, and direct enumeration gives extension values 13 and 15. Thus even the entire scalar free-energy curve loses the geometry required by a Bellman step. This does not refute a nonlocal minimax interpolation with a richer order parameter.

The verifier also exhausts all root-normalized residual graphs through order eight. If $\mathcal{B}_n$ is the coordinatewise Pareto-minimal set of pairs $(M(B), \delta_w(B))$, then

n	$\mathcal{B}_n$
2	$\{(1, 1)\}$
3	$\{(3, 0)\}$
4	$\{(4, 0)\}$
5	$\{(4, 0)\}$
6	$\{(5, 2), (7, 1), (9, 0)\}$
7	$\{(9, 0)\}$
8	$\{(10, 1), (12, 0)\}$.

Every point at order six gives $F(7) = 9$, and both points at order eight give $F(9) = 12$. Thus two units of additional internal energy can be exactly offset by one unit of covering-deficit improvement. This is finite evidence for a near-optimal Pareto family, not an asymptotic conclusion.

9.1 The nonlinear square-covariance gain

The earlier Gaussian proof used only the inequality $\arcsin'(u) \geq 1$. Retaining the discarded nonlinear term gives a stability theorem.

Theorem 13 (Nonlinear Gaussian-sign bound). *For $s \geq 0$, put*

$$D_s = 1 + s^2(n-1), \quad d_s = \frac{2s}{D_s}, \quad z_{ij,s} = \frac{s^2|(A^2)_{ij}|}{D_s}.$$

Then every order- n signing satisfies

$$M(A) \geq \frac{1}{\pi} \sum_{i < j} [\arcsin(z_{ij,s} + d_s) - \arcsin(z_{ij,s} - d_s)]. \quad (25)$$

Consequently

$$M(A) \geq \frac{n(n-1)}{\pi} \arcsin \frac{1}{\sqrt{n-1}}, \quad (26)$$

and, for $n \geq 3$,

$$M(A) \geq \frac{n(n-1)}{\pi} \arcsin \frac{1}{\sqrt{n-1}} + \frac{\text{tr}(A^4) - n(n-1)^2}{8\pi(n-1)(n-2)^{3/2}}. \quad (27)$$

Proof. The two matrices

$$R_s^\pm = \frac{(I \pm sA)^2}{D_s}$$

are correlation matrices. With $u_{ij} = s^2 a_{ij}(A^2)_{ij}/D_s$, the corresponding Gaussian sign roundings and the arcsine law give

$$\mathbb{E}Q_A(X^+) - \mathbb{E}Q_A(X^-) = \frac{2}{\pi} \sum_{i < j} [\arcsin(u_{ij} + d_s) - \arcsin(u_{ij} - d_s)].$$

The bracket is even in u_{ij} , and each expectation belongs to $[-M(A), M(A)]$, proving (25).

For $f_d(z) = \arcsin(z + d) - \arcsin(z - d)$ and $z \geq 0$,

$$f_d''(z) = h(z + d) - h(z - d), \quad h(t) = \frac{t}{(1 - t^2)^{3/2}}.$$

Thus f_d is even and convex. At $s = 1/\sqrt{n-1}$ one has $D_s = 2$ and $d_s = 1/\sqrt{n-1}$, so $f_d(z) \geq f_d(0) = 2 \arcsin d$, which proves (26).

Write $q = n - 1$. Since

$$h'(t) = \frac{1 + 2t^2}{(1 - t^2)^{5/2}}$$

is even and increasing in $|t|$, $f_d''(z)$ is increasing for $z \geq 0$. For $d = 1/\sqrt{q}$,

$$f_d(z) \geq 2 \arcsin d + \frac{q}{(q-1)^{3/2}} z^2.$$

Finally $z_{ij} = |(A^2)_{ij}|/(2q)$ and

$$\sum_{i < j} z_{ij}^2 = \frac{\text{tr}(A^4) - nq^2}{8q^2},$$

which proves (27). □

The excess is exactly

$$\text{tr}(A^4) - n(n-1)^2 = \|A^2 - (n-1)I\|_F^2.$$

Hence a sequence attaining the asymptotic lower constant $1/\pi$ must obey $\|A_n^2 - (n-1)I\|_F = o(n^2)$. The universal improvement in (26) is only $\Theta(\sqrt{n})$, so it does not change the leading constant. It does imply the new exact consequence

$$\boxed{F(21) \geq 32}.$$

Indeed (26) is greater than 30 and all order-21 energies are even. For an exact strictness certificate, use $\pi/14 < 11/49$ and $\sin t < t - t^3/6 + t^5/120$; rational squaring shows that the latter polynomial at $11/49$ has square less than $1/20$. Thus $\arcsin(1/\sqrt{20}) > \pi/14$.

9.2 The multivertex Bellman state and its entropy bound

For internal signings B, D of orders n, k , define

$$J(B, D) = \min_C M \begin{pmatrix} B & C \\ C^\top & D \end{pmatrix}, \quad L(B, D) = \max_{x, y} |Q_B(x) + Q_D(y)|.$$

In the projective Hamming cube $\mathcal{P}_{nk} = \{\pm 1\}^{n \times k} / \{C \sim -C\}$, let

$$\mathcal{R}_{n,k} = \{[xy^\top] : x \in \{-1, 1\}^n, y \in \{-1, 1\}^k\}$$

and give $R = [xy^\top]$ the integer weight

$$w_{B,D}(R) = \frac{L(B, D) - |Q_B(x) + Q_D(y)|}{2}.$$

Put

$$\rho_w^\square(B, D) = \max_{[C] \in \mathcal{P}_{nk}} \min_{R \in \mathcal{R}_{n,k}} [d_\pm([C], R) + w_{B,D}(R)].$$

Theorem 14 (Multivertex weighted Bellman identity). *One has*

$$J(B, D) = L(B, D) + nk - 2\rho_w^\square(B, D). \quad (28)$$

Thus, with

$$\delta_w^\square = \left\lfloor \frac{nk}{2} \right\rfloor - \rho_w^\square,$$

$$F(n+k) = \min_{B,D} [L(B, D) + (nk \bmod 2) + 2\delta_w^\square(B, D)]. \quad (29)$$

Proof. Pairing y with $-y$ gives

$$\max_{\eta=\pm 1} |Q_B(x) + Q_D(y) + \eta x^\top C y| = |Q_B(x) + Q_D(y)| + |x^\top C y|.$$

Moreover $|x^\top C y| = nk - 2d_\pm([C], [xy^\top])$. Substitute $|Q_B + Q_D| = L - 2w$, maximize over the rank-one code, and minimize over $[C]$ to obtain (28). A weight-zero rank-one word is within projective distance $\lfloor nk/2 \rfloor$ of every center, so the deficit is nonnegative. Minimizing over the two internal blocks proves (29). $\square$

Only the internal window

$$|Q_B(x) + Q_D(y)| \geq L(B, D) - 2 \left\lfloor \frac{nk}{2} \right\rfloor$$

can affect this radius. Define its soft effective size by

$$\Xi(B, D) = \log \sum_{R \in \mathcal{R}_{n,k}} \exp \left(-\frac{2w_{B,D}(R)^2}{nk} \right).$$

Proposition 15 (Weighted entropy upper bound).

$$J(B, D) \leq L(B, D) + \sqrt{2nk(\Xi(B, D) + \log 4)}. \quad (30)$$

Proof. Choose C uniformly at random. Hoeffding's inequality and a union bound give

$$\Pr\{\exists R : |\langle C, R \rangle| > T + 2w(R)\} \leq 2 \exp \left(-\frac{T^2}{2nk} + \Xi(B, D) \right).$$

For $T = \sqrt{2nk(\Xi + \log 4)}$ this is at most $1/2$. Hence some C satisfies every displayed inequality, and $L - 2w(R) + |\langle C, R \rangle| \leq L + T$ for every R . Theorem 14 finishes the proof. $\square$

The generic bound $\Xi \leq (n + k - 2) \log 2$ reproduces the known leading $\sqrt{nk(n+k)}$ block cost. A sufficient replacement is a constant-matching profile law: find B, D and $\varepsilon > 0$ such that

$$L(B, D) + \sqrt{2nk(\Xi(B, D) + \log 4)} \leq (F(n)^{2/3} + F(k)^{2/3})^{3/2} + O((n+k)^{3/2-\varepsilon}). \quad (31)$$

This would imply power-saving near-subadditivity of $F^{2/3}$, which the standard blocking argument turns into convergence; (31) is a target, not a proved estimate.

There is also density-one control of the exact one-vertex state. Choose B_n attaining Theorem 12 and put $\sigma_n = M(B_n) - F(n)$ and $\delta_n = \delta_w(B_n)$. Exactly

$$F(n+1) - F(n) = \sigma_n + (n \bmod 2) + 2\delta_n.$$

Telescoping and $F(n) = O(n^{3/2})$ show

$$\sum_{n=N}^{2N} (\sigma_n + 2\delta_n) = O(N^{3/2}).$$

Thus, for every $g(N) \rightarrow \infty$, all but $O(N/g(N))$ indices in $[N, 2N]$ have a Bellman-optimal predecessor with $\sigma_n \leq g(N)\sqrt{N}$ and $\delta_n \leq \frac{1}{2}g(N)\sqrt{N}$. Sparse exceptional orders remain uncontrolled, so this does not prove convergence.

9.3 Complete order-nine weighted geometry

Exhausting all 12,346 root-normalized order-nine signings gives the new full frontier

$$\mathcal{B}_9 = \{(12, 0)\}.$$

The 55 root records with $M = 12$ split as 20 with deficit zero and 35 with deficit one; their 15 switching-permutation classes split as four and eleven.

The records `GH0gmo` and `Gxd?Dc` give a stronger collision. Both have $M = 14$, projective energy histogram

$$\#\{|Q| = 2, 6, 10, 14\} = (124, 85, 37, 10),$$

and ordered pair-distance enumerator $(10, 16, 14, 20, 40)$ for their ten exact maximizers. Their profiles $(\rho_{\text{ext}}, \rho_w, E)$ are, however, $(4, 4, 15)$ and $(3, 3, 17)$. Hence the energy histogram plus the two-point overlap law of exact maximizers still does not determine extension.

Across the full rooted catalogue, ten histogram groups containing 874 records mix deficits. Adding the maximizer pair-distance law leaves three mixed groups containing 112 records. The complete energy-coloured two-point law

$$T_B(e, f, r) = \#\{([x], [y]) : |Q_B(x)| = e, |Q_B(y)| = f, d_{\pm}([x], [y]) = r\}$$

separates all deficits at order nine. This is a finite computation, not an asymptotic sufficiency claim. The verifier asserts the nauty stream count and digest and recomputes every extension both directly and from the weighted radius.

10 Six proof programs and the sharp remaining lemma

The detailed derivations and failed variants are preserved in `RESEARCH_LEDGER.md`. The current programs are logically distinct.

A. Quenched extensive free energy. Prove $\lim_n \Psi_{n,\beta}$ for every fixed $\beta > 0$, or at least for an unbounded set of β 's. Equation (4) then forces convergence. Proposition 4 shows exactly why independent annealed cross blocks fail. A repair must absorb the cross term in a joint minimax order parameter, as random-disorder spin-glass interpolation absorbs it only after disorder expectation [3, 4].

B. Near-saturated relaxation. Find a nonconvex compact relaxation whose optimizers have $\sum_e (1 - a_e^2) = o(n^2)$ and an asymptotic composition law. Then Proposition 3 transfers its limit to signs. The cube and sphere calculations prove that the two obvious feasible sets are structurally too loose or exactly discrete.

C. Signed Eulerian polynomial. Prove a second-order limit for $\min_y \log P_y(t)$ after the explicit MacWilliams terms are included. Puncturing and ordinary products fail because of the common parity phase, rank-one cross lift, and macroscopic family of mixed 4-cycles. A quenched coset-weight theorem, rather than an average weight enumerator, is required.

D. Cavity covering profile. Control the Pareto set $(M(B), \delta_w(B))$ uniformly in the near-optimal energy window. Theorem 12 makes this state exact for one vertex, while the order-seven computation rules out the smaller state consisting only of exact maximizers. A useful theorem must retain enough energy layers to control the weighted radius without paying a leading-order entropy loss.

E. Second-order limit object. Construct a rate-sensitive topology for $A/\sqrt{n}$ in which the Boolean maximum is continuous and sign arrays have an asymptotically lossless composition. Theorem 9 and Proposition 10 rule out ordinary graphons and empirical spectra. Spectral edges, eigenvectors, or the whole induced Gaussian process would have to remain visible.

F. Nonexistence. To separate subsequences one needs order-specific lower bounds on $F(n)$, not merely constructions with different energies. Conference, random, and planted matrices supply upper bounds or deliberately inferior examples. The small values fluctuate but do not certify a subsequence mechanism.

No route presently closes. The single best next lemma is the following.

Quenched minimax free-energy lemma. For every fixed $\beta > 0$, the limit

$$\lim_{n \rightarrow \infty} \min_A \frac{1}{\beta n} \log \sum_{x,s} \exp \left(\frac{\beta s Q_A(x)}{\sqrt{n}} \right)$$

exists.

Proving this lemma for an unbounded collection of inverse temperatures settles the MathOverflow question by (4). In coding coordinates it asks for cancellation in $P_y(\tanh(\beta/\sqrt{n}))$. Any proof that bounds the mixed cross-block contribution independently incurs a leading error; a successful proof must make that contribution part of the optimized variational structure.

Reproducibility and final status

The exact identity checker, exhaustive small-order search, optional Z3 certificate, expected outputs, deterministic seeds, and certificate boundaries are documented in `verification/README.md`. The

literature map is in `LITERATURE.md`; the original and second proof-search ledgers are both preserved.

The adversarial ellipsope limit, the code dual and MacWilliams formula, the oriented covariance barrier, the graph-limit obstructions, finite non-heredity, and the weighted covering-radius Bellman identity materially narrow the search. They do not prove existence or nonexistence of the original limit.

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